

Loan Default Analysis: A Combined Python and SQL Approach

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Executive Summary

This project implements a hybrid analytical pipeline using Python for data preprocessing and SQL Server for structured exploratory data analysis. The dataset contains **255,347 borrower records**, providing a large sample size for segmentation and pattern analysis.

The system successfully performs multi-dimensional segmentation (income, credit score, DTI, employment, and loan purpose). However, a **critical failure in the target variable (Defaulter)** prevents meaningful risk analysis. All records show non-informative or zero values, making default-based insights invalid.

Despite this limitation, the project demonstrates strong analytical design, proper data engineering practices, and the ability to identify real-world data quality issues — a key skill in data analytics.

1. System Architecture and Objective

Architecture

Layer	Technology	Role
Data Preparation	Python	Cleaning, transformation, feature engineering
Analysis	SQL Server	Segmentation, aggregation, insight extraction

Objective

The system is designed to:

- Transform raw loan data into an analysis-ready format
 - Perform structured segmentation using SQL
 - Analyze borrower distribution across financial and demographic variables
 - Evaluate whether the dataset supports meaningful default risk analysis
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2. Python: The Data Preparation Layer

Python plays a foundational role by converting raw, inconsistent data into a structured dataset.

2.1 Data Cleaning

The dataset initially contained:

- Missing values in multiple columns (e.g., LoanTerm)
- Mixed data types (numeric stored as text)
- Inconsistent categorical values

These were resolved by:

- Handling null values
- Converting columns into numeric formats
- Standardizing categorical entries

Impact:

This step ensures SQL queries operate on consistent and reliable data.

2.2 Feature Engineering

To improve interpretability, new derived variables were created:

Income Groups

- Low Income (< 40,000)
- Middle Income (40,000–80,000)
- High Income (> 80,000)

Fact:

- ~177,000 borrowers fall into the **high-income category (~69%)**

- Indicates a strong dataset skew
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Credit Score Categories

- Poor (< 580)
- Fair (580–669)
- Good (670–739)
- Excellent (740+)

Fact:

- Actual values observed are around **~50–70 range (scaled)**
 - Indicates normalization or transformation
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DTI Ratio Segmentation

- Low Risk (< 0.2)
- Medium Risk (0.2–0.4)
- High Risk (> 0.4)

Fact:

- Majority of records fall into **high DTI category (> 0.4)**

2.3 Data Transition

- Cleaned dataset exported as CSV
- Imported into SQL Server (`loan_default` table)

This enables efficient querying on large-scale structured data.

3. SQL: The Analysis Layer

SQL is used to perform large-scale aggregation and segmentation.

3.1 Data Preparation in SQL

- `Default` column converted to numeric (1/0)
 - Used `TRY_CAST` to handle invalid values
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3.2 Analytical Findings

A. Borrower Distribution

Category	Observation	Interpretation
Employment Type	~85k in each category	Uniform distribution
Marital Status	Almost equal across groups	No demographic dominance
Loan Purpose	Nearly 50-50 split	Balanced dataset

Insight:

Dataset is **well-balanced structurally**, which is good for analysis.

B. Income Segmentation

Category	Approx Count
High Income	~177,000

Middle Income ~41,000

Low Income ~36,000

Insight:

- Strong skew toward high-income borrowers
- Low-income risk patterns cannot be properly analyzed

C. Credit Score Analysis

Observation:

- CreditScore values are around **~59 average**

Interpretation:

- Dataset uses **scaled/normalized values**
- Standard credit thresholds (300–850) cannot be applied

D. DTI Ratio Analysis

Observation:

- Majority of borrowers fall into **high DTI category**

Interpretation:

- Either:
 - Incorrect threshold selection
 - Improper scaling

Impact:

- DTI loses its effectiveness as a risk indicator

E. Co-signer Analysis

Category	Count
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No Co-signer	~225,000
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With Co-signer ~29,000

Insight:

- ~88% borrowers are independent
- Co-signer impact is minimal

F. Multi-Factor Segmentation

- Combined EmploymentType + LoanPurpose

Observation:

- Segmentation works technically
 - No risk differentiation visible
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4. Critical Problem and Limitations

Critical Problem: Defaulter Column

Issue

- All values = 0 or invalid
- No variation

Fact

- Total Defaulters = 0 across **255,347 rows**

Impact

- Default rate = 0% everywhere
 - Risk modeling impossible
 - No predictive insights possible
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Root Cause Analysis

Likely reasons:

- Incorrect CSV export/import
 - Improper data type conversion
 - Loss of categorical meaning in Python
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System Limitations

- No usable target variable
 - CreditScore improperly scaled
 - DTI distribution unrealistic
 - Missing values (LoanTerm) reduce completeness
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5. Recommendations

To make the system usable:

1. Fix Target Variable

- Reprocess `Defaulter` column in Python
- Ensure binary encoding (1/0) is correct

2. Validate Scaling

- Recheck CreditScore range
- Adjust DTI thresholds

3. Handle Missing Data

- Impute or clean LoanTerm

4. Validate Pipeline

- Ensure no data loss during CSV export/import
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6. Conclusion and Key Learning

This project successfully demonstrates a **real-world data analysis pipeline**:

- Python for data preparation
- SQL for structured analysis
- Segmentation and aggregation logic

However, the most important takeaway is:

Even a perfectly designed system fails if the data is incorrect.

Key Learning

- Data validation is as important as analysis
- Detecting data issues is a core analyst skill
- Real-world datasets often contain hidden flaws